

Learning Activity 10: Raking for Nonresponse Adjustment

STA304

Goal

Raking is a weighting adjustment method used when we know population margins for two or more variables, but we do not know the full population counts for every cell in the post-stratification.

The outcome of interest is the total number of weekly study hours. You suspect that women are less likely to answer this than men question and that the year of the study will also affect response propensities. You worry because you suspect both of these factors (sex and year of study) are also correlated with the total number of weekly study hours.

The number of females in the population is 60, the number of males in the target population is 40. The number of first years, second years, and third years in the target population are 25, 35, and 40. There are 100 people in the population.

```
set.seed(2)
srs_sample <- population[sample(1:100, 50, replace = FALSE),]

srs_sample %>% ungroup() %>% select(sex, year, obs_hours) %>% head(10)
```

```
##      sex year obs_hours
## 85   Male  3rd    14.4
## 79   Male  3rd    13.8
## 70   Male  2nd    15.5
## 6    Male  2nd    13.2
## 32   Male  2nd    14.8
## 8    Female 3rd     NA
## 17 Female  3rd     NA
## 93   Male  1st    15.3
## 81 Female  3rd     NA
## 76 Female  3rd     NA
```

Instructor Solution Key with R

The code below calculates the exact numerical solutions from the simulated sample used in this activity.

```
# Targets
row_targets <- c(Female = 60, Male = 40)
col_targets <- c("1st" = 25, "2nd" = 35, "3rd" = 40)
```

```

# Respondent data only
respondents <- srs_sample %>%
  ungroup() %>%
  filter(!is.na(obs_hours))

# Initial SRS design weight
initial_weight <- N / 50
initial_weight

```

```
## [1] 2
```

Part 1 Solutions

a)

The initial survey weight is

$$w_i = \frac{N}{n} = \frac{100}{50} = 2.$$

So every sampled respondent has initial weight 2.

b)

```

# Respondent counts by sex and year
respondent_counts <- xtabs(~ sex + year, data = respondents)
respondent_counts

```

```
##           year
## sex      1st 2nd 3rd
## Female    6  2  4
## Male      6  5  6
```

```

# Initial weighted respondent table
w0 <- respondent_counts * initial_weight
w0

```

```
##           year
## sex      1st 2nd 3rd
## Female   12  4  8
## Male     12 10 12
```

```

# Add row and column totals
addmargins(w0)

```

```
##           year
## sex      1st 2nd 3rd Sum
##  Female   12  4   8  24
##   Male    12 10  12  34
##    Sum     24 14  20  58
```

Part 2 Solutions

a)

```
rowSums(w0)
```

```
## Female   Male
##      24     34
```

```
row_targets
```

```
## Female   Male
##      60     40
```

```
colSums(w0)
```

```
## 1st 2nd 3rd
##  24  14  20
```

```
col_targets
```

```
## 1st 2nd 3rd
##  25  35  40
```

The initial respondent table does not match the known sex margins or the known year margins.

b)

We cannot simply post-stratify because we only know the marginal totals for sex and year. We do not know the full population counts for every sex $\times$ year cell.

Part 3 Solutions: First Row Adjustment

```
# Row adjustment factors
row_factors_1 <- row_targets / rowSums(w0)
row_factors_1
```

```
##      Female      Male
## 2.500000 1.176471
```

```
# Apply row adjustment
w1 <- sweep(w0, 1, row_factors_1, FUN = "*")
w1
```

```
##           year
## sex        1st      2nd      3rd
## Female 30.00000 10.00000 20.00000
## Male   14.11765 11.76471 14.11765
```

```
# Check margins
addmargins(w1)
```

```
##           year
## sex        1st      2nd      3rd      Sum
## Female 30.00000 10.00000 20.00000 60.00000
## Male   14.11765 11.76471 14.11765 40.00000
## Sum    44.11765 21.76471 34.11765 100.00000
```

After this step, the row totals match the known sex totals exactly, but the column totals do not match the known year totals.

Part 4 Solutions: First Column Adjustment

```
# Column adjustment factors based on the table after row adjustment
col_factors_1 <- col_targets / colSums(w1)
col_factors_1
```

```
##           1st      2nd      3rd
## 0.5666667 1.6081081 1.1724138
```

```
# Apply column adjustment
w2 <- sweep(w1, 2, col_factors_1, FUN = "*")
w2
```

```
##           year
## sex        1st      2nd      3rd
## Female 17.00000 16.08108 23.44828
## Male    8.00000 18.91892 16.55172
```

```
# Check margins
addmargins(w2)
```

```
##           year
## sex      1st      2nd      3rd      Sum
## Female 17.00000 16.08108 23.44828 56.52936
## Male   8.00000 18.91892 16.55172 43.47064
## Sum    25.00000 35.00000 40.00000 100.00000
```

After this step, the column totals match the known year totals exactly, but the row totals do not still match the known sex totals.

Part 5 Solutions: Second Row Adjustment

```
# Second row adjustment factors
row_factors_2 <- row_targets / rowSums(w2)
row_factors_2
```

```
##      Female      Male
## 1.0613954 0.9201612
```

```
# Apply second row adjustment
w3 <- sweep(w2, 1, row_factors_2, FUN = "*")
w3
```

```
##           year
## sex      1st      2nd      3rd
## Female 18.04372 17.06839 24.88789
## Male   7.36129 17.40846 15.23025
```

```
# Check margins
addmargins(w3)
```

```
##           year
## sex      1st      2nd      3rd      Sum
## Female 18.04372 17.06839 24.88789 60.00000
## Male   7.36129 17.40846 15.23025 40.00000
## Sum    25.40501 34.47684 40.11815 100.00000
```

The row totals again match the known sex totals. The column totals are closer to the known year totals, but they may not match exactly yet.

Part 6 Solutions: Raking Until Convergence

```

# Start from the initial weighted respondent table
raked <- w0

max_iter <- 100
tol <- 1e-6

for (iter in 1:max_iter) {

  # Step 1: adjust rows
  row_factors <- row_targets / rowSums(raked)
  raked <- sweep(raked, 1, row_factors, FUN = "*")

  # Step 2: adjust columns
  col_factors <- col_targets / colSums(raked)
  raked <- sweep(raked, 2, col_factors, FUN = "*")

  # Check convergence
  row_error <- max(abs(rowSums(raked) - row_targets))
  col_error <- max(abs(colSums(raked) - col_targets))
  max_error <- max(row_error, col_error)

  if (max_error < tol) {
    break
  }
}

iter

```

```
## [1] 6
```

```
max_error
```

```
## [1] 7.570083e-08
```

```
raked
```

```
##           year
## sex      1st      2nd      3rd
## Female 17.778552 17.365660 24.855788
## Male   7.221448 17.634340 15.144212
```

```
rowSums(raked)
```

```
## Female  Male
##      60    40
```

```
colSums(raked)
```

```
## 1st 2nd 3rd
## 25 35 40
```

```
addmargins(raked)
```

```
##          year
## sex          1st          2nd          3rd          Sum
## Female 17.778552 17.365660 24.855788 60.000000
## Male   7.221448 17.634340 15.144212 40.000000
## Sum    25.000000 35.000000 40.000000 100.000000
```

Part 7 Solutions

a)

In this context, `sweep()` multiplies each row or column of the table by its corresponding adjustment factor.

For example,

```
sweep(w0, 1, row_factors_1, FUN = "*")
```

```
##          year
## sex          1st          2nd          3rd
## Female 30.000000 10.000000 20.000000
## Male   14.11765 11.76471 14.11765
```

multiplies each row of `w0` by the appropriate row adjustment factor.

b)

The algorithm converges when the row totals and column totals are both very close to their target population margins. Specifically, “closeness” was measured as the max absolute difference between all rows and column and their theoretical totals.

c)

The tolerance value controls how close the weighted margins need to be to the target margins before the algorithm stops.

```
tol
```

```
## [1] 1e-06
```

```
max_error
```

```
## [1] 7.570083e-08
```

```
max_error < tol
```

```
## [1] TRUE
```

Part 8 Solutions: Estimating a Mean After Raking

a)

First calculate the observed mean number of study hours in each sex $\times$ year cell.

```
cell_means <- tapply(
  respondents$obs_hours,
  list(respondents$sex, respondents$year),
  mean
)

cell_means
```

```
##           1st    2nd    3rd
## Female 13.16667 17.90 23.05
## Male   9.15000 14.92 19.35
```

The raked estimated total is

$$\hat{t}_{rake} = \sum_h w_h^* \bar{y}_h,$$

where w_h^* is the final raked weight total for cell h , and $\bar{y}_h$ is the observed mean in that cell.

```
raked_total <- sum(raked * cell_means)
raked_total
```

```
## [1] 1740.077
```

The respondent-only SRS estimate, ignoring nonresponse, is

$$\hat{t}_{resp} = N \bar{y}_R.$$

```
respondent_only_total <- N * mean(respondents$obs_hours)
respondent_only_total
```

```
## [1] 1560.69
```

```
true_total <- sum(population$true_hours)
true_total
```

```
## [1] 1715.8
```

```
comparison_total <- data.frame(
  estimate = c("Respondent-only estimate", "Raked estimate", "True population total"),
  total = c(respondent_only_total, raked_total, true_total),
  error = c(respondent_only_total - true_total,
            raked_total - true_total,
            0)
)
```

```
comparison_total
```

```
##           estimate    total    error
## 1 Respondent-only estimate 1560.690 -155.1103
## 2           Raked estimate 1740.077   24.2766
## 3    True population total 1715.800    0.0000
```

b)

The raked estimate of the population mean is

$$\hat{Y}_{rake} = \frac{\hat{t}_{rake}}{N}.$$

```
raked_mean <- raked_total / N
respondent_only_mean <- mean(respondents$obs_hours)
true_mean <- mean(population$true_hours)
```

```
comparison_mean <- data.frame(
  estimate = c("Respondent-only mean", "Raked mean", "True population mean"),
  mean_hours = c(respondent_only_mean, raked_mean, true_mean),
  error = c(respondent_only_mean - true_mean,
            raked_mean - true_mean,
            0)
)
```

```
comparison_mean
```

```
##           estimate mean_hours    error
## 1 Respondent-only mean   15.60690 -1.551103
## 2           Raked mean   17.40077  0.242766
## 3 True population mean   17.15800  0.000000
```

The raked estimate is closer to the truth in this example because response depends on sex and year, and both variables are related to study hours.